LAB 4: Triggers

The assignment:

1. Write a PL/SQL program to create smart autoincrement function in the table defined below

```
Create table table_without_autoincrement(
        id number,
        first_name varchar2(40),
        last_name varchar2(40));
```

a) When a new row is added to table table_without_autoincrement and id is set (and here is no row with such value) trigger will add such unchanged row,
b) Whew a new row is added to table table_without_autoincrement and id of a new row is null, trigger will change null value with the first available value, i.e. if existing IDs in table_without_autoincrement constitute sequence without gaps, trigger will replace null value in a new row with value equal to max(ID)+1, in other case trigger will use the smallest value from the first gap.

The code:

Worksheet Query Builder

```
CREATE OR REPLACE TRIGGER trg_smart_autoincrement
BEFORE INSERT ON table_without_autoincrement
FOR EACH ROW
DECLARE
   v_id NUMBER := 1;
   v_count NUMBER;
BEGIN

   -- b) ID is NULL
   -- Find the first available ID and assign it
   IF :NEW.id IS NULL THEN

      LOOP
         SELECT COUNT(*)
         INTO v_count
         FROM table_without_autoincrement
         WHERE id = v_id;

         EXIT WHEN v_count = 0;

         v_id := v_id + 1;
      END LOOP;

      :NEW.id := v_id;
```

```
    ELSE

        -- a) ID is provided by the user
        -- Check if such ID already exists
        SELECT COUNT(*)
        INTO v_count
        FROM table_without_autoincrement
        WHERE id = :NEW.id;

        IF v_count > 0 THEN
            RAISE_APPLICATION_ERROR(
                -20001,
                'ID already exists'
            );
        END IF;

        -- If v_count = 0, the row is inserted unchanged

    END IF;

END;
/
```

Test a (ID provided):

```
INSERT INTO table_without_autoincrement
VALUES (5, 'Karla', 'Budimir');
```

It works.

```
INSERT INTO table_without_autoincrement
VALUES (5, 'Josip', 'Bosnic');
```

It raises error:

```
Error report -
ORA-20001: ID already exists
ORA-06512: at "STUDENT.TRG_SMART_AUTOINCREMENT", line 33
ORA-04088: error during execution of trigger 'STUDENT.TRG_SMART_AUTOINCREMENT'
```

Test b (ID NULL):

```
INSERT INTO table_without_autoincrement
VALUES (NULL, 'Veronika', 'Budimir');

Trigger automatically assigns the first available ID.
```

To verify:

```
SELECT *
FROM table_without_autoincrement
ORDER BY id;
```

ID	FIRST_NAME	LAST_NAME
1	Veronika	Budimir
5	Karla	Budimir

1 is the smallest available ID so that one is chosen.